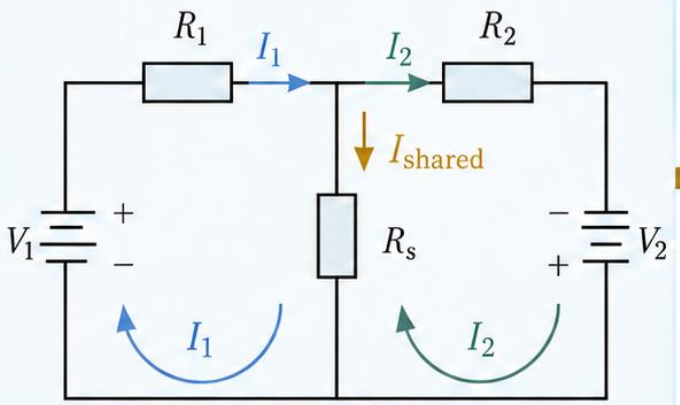
Linear Systems: Assemble and validate two-loop Kirchhoff equations

Two-loop DC circuit



Kirchhoff equations

One KVL equation for each mesh:

Eq. 1: . . .

Eq. 2: . . .

Matrix form

$$A\,x = b$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Stores the two equations compactly.

Solve and validate

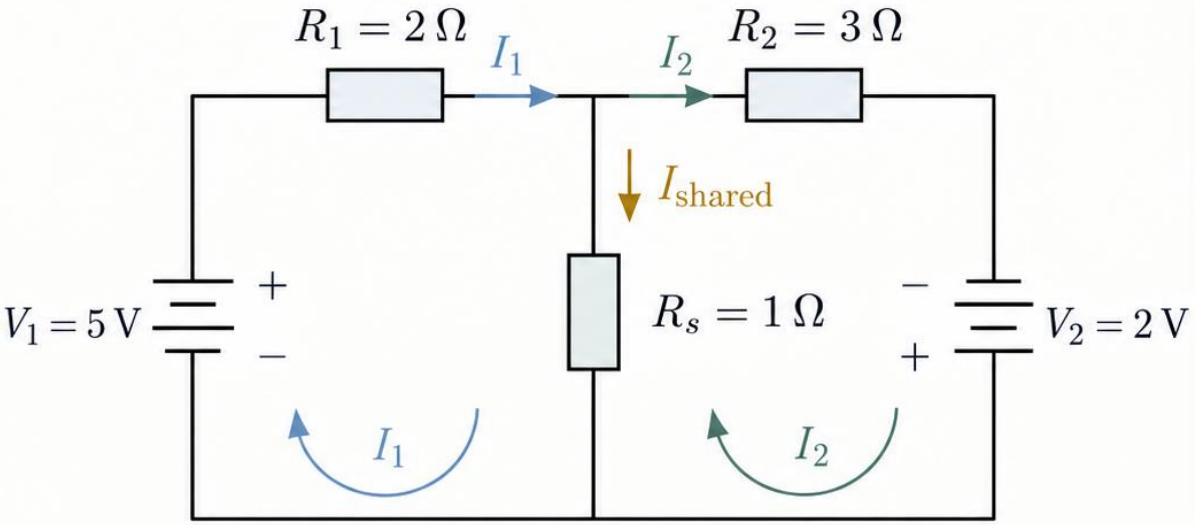
```
x = A \ b;
I1 = x(1);
I2 = x(2);
```

Check the result against the original equations.

Key points

- A two-loop circuit can be written as **two equations**.
- A **2x2 system** stores them compactly.
- **MATLAB** can solve the system.
- The result still needs a **physical check**.

Choose current directions at the junction before writing equations



SIGN CONVENTIONS & PREDICTIONS

- Choose clockwise mesh currents I_1 and I_2.
- At the junction: I_1 enters; I_2 and I_{shared} leave.
- $I_1 = I_2 + I_{\text{shared}}$
- $I_{\text{shared}} = I_1 - I_2$
- Prediction: $I_1 > I_2$

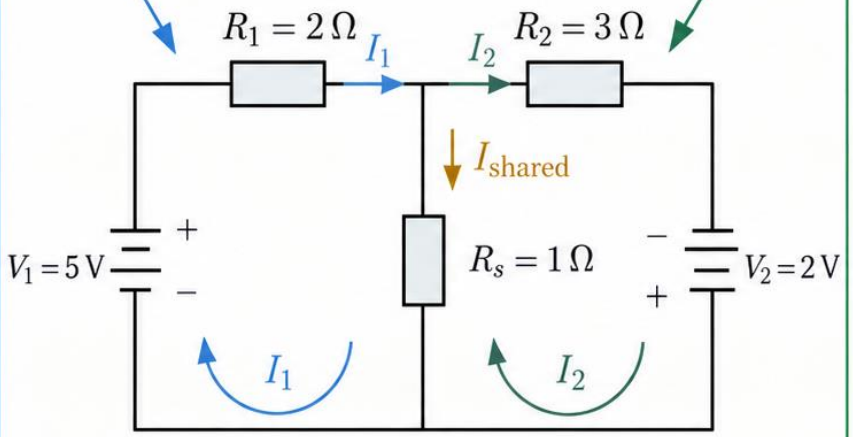
Before algebra: which current direction will the shared branch take?

Walk each loop in words before writing symbols

LEFT LOOP WALK

1. Enter through $V_1 = 5\ \text{V}$.
2. Drop across R_1 and the shared branch.
3. Shared contribution follows $I_1 - I_2$.

$$(R_1 + R_s)I_1 - R_sI_2 = V_1$$



RIGHT LOOP WALK

1. Enter through $V_2 = 2\ \text{V}$ with its shown polarity.
2. Drop across R_2 and traverse the shared branch in reverse.
3. Shared contribution is reversed.

$$-R_sI_1 + (R_2 + R_s)I_2 = V_2$$

One physical loop → one Kirchhoff equation

Every resistance-current term has volts: Ω × A = V

Turn each Kirchhoff equation into one matrix row

Left loop (row 1)

KVL (clockwise):

$$(R_1 + R_s)\,I_1 - R_s\,I_2 = V_1$$

$$(2+1)\,I_1 - 1\,I_2 = 5$$

$$3\,I_1 - 1\,I_2 = 5$$

Right loop (row 2)

KVL (clockwise):

$$-R_s\,I_1 + (R_2 + R_s)\,I_2 = V_2$$

$$-1\,I_1 + (3+1)\,I_2 = 2$$

$$-1\,I_1 + 4\,I_2 = 2$$

$A\,\mathbf{x} = \mathbf{b}$

$$\underbrace{\begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix}}_{A\ (\Omega)} \underbrace{\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}}_{\mathbf{x}\ (\mathrm{A})} = \underbrace{\begin{bmatrix} 5 \\ 2 \end{bmatrix}}_{\mathbf{b}\ (\mathrm{V})}$$

Unknown order and source order:

- $\mathbf{x} = [I_1; I_2]$.
- Row 1 ← left loop, Row 2 ← right loop.
- $\mathbf{b} = [V_1; V_2]$.

KEY POINTS

- Keep $\mathbf{x} = \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$.
- Row 1 comes from the left loop.
- Row 2 comes from the right loop.
- $\mathbf{b}$ stores matching source terms.

Keep the units visible in A*x=b

$$\begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix} \quad \Omega \times \mathrm{A} = \mathrm{V}$$

Ω

$A = [3\ -1; -1\ 4]$ ohm.

A

$x = \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$ A.

V

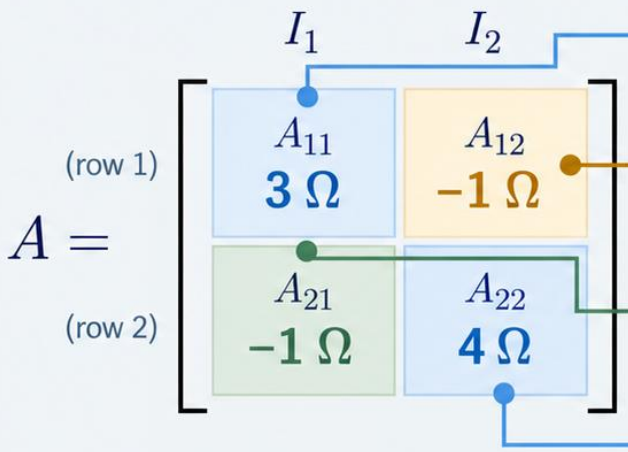
$b = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$ V.

V

A*x has volts.

Audit the matrix before asking MATLAB to solve

Coefficient matrix *A*



Entry	Value	Physical origin
A_{11}	$3\,\Omega$	$R_1 + R_s$: resistance multiplying I_1 in the left-loop equation
A_{12}	$-1\,\Omega$	$-R_s$: effect of I_2 on the shared-resistor drop in the left loop
A_{21}	$-1\,\Omega$	$-R_s$: effect of I_1 on the shared-resistor drop in the right loop
A_{22}	$4\,\Omega$	$R_2 + R_s$: resistance multiplying I_2 in the right-loop equation

Every coefficient should read back as physics.

Check the value, the sign, and the physical origin of each matrix entry before solving in MATLAB.

Solve the supplied system with MATLAB backslash

MATLAB code (supplied)

```
R1_ohm = 2; R2_ohm = 3; Rs_ohm = 1;
V1_V = 5; V2_V = 2;
A_ohm = [R1_ohm+Rs_ohm, -Rs_ohm; ...
         -Rs_ohm, R2_ohm+Rs_ohm];
b_V = [V1_V; V2_V];
x_A = A_ohm\b_V;
I1_A = x_A(1);
I2_A = x_A(2);
```

Store resistances and source voltages

Coefficient matrix *A*

Source vector b

Backslash solve

Mesh currents *x*

Result (from supplied code)

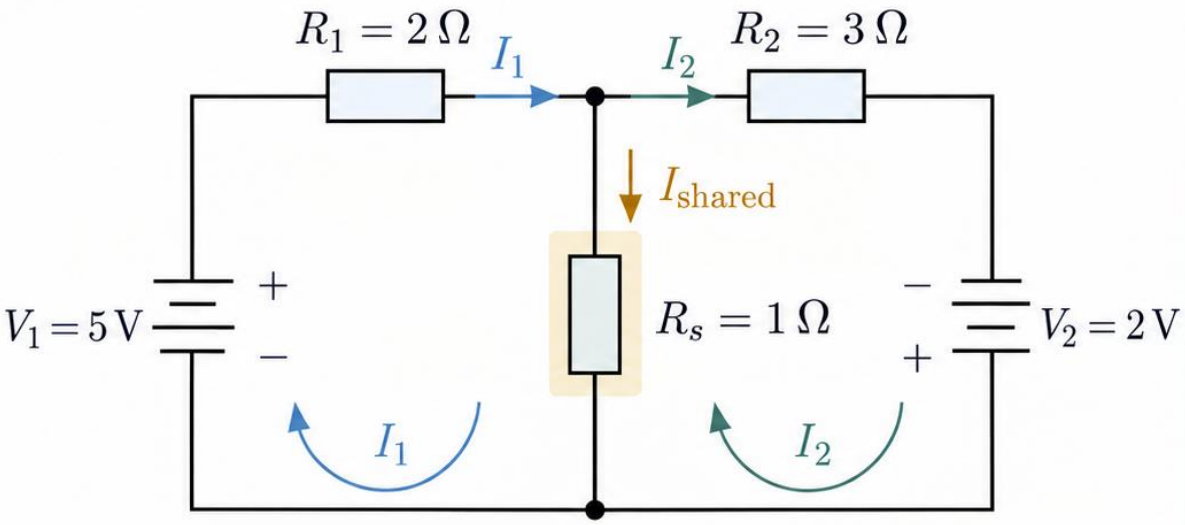
```
I1_A = 2 A
I2_A = 1 A
```

KEY STEPS

- Store resistances and source voltages.
- Build A and b in order.
- Use `x_A = A_ohm\b_V`.
- Read `x_A(1)` and `x_A(2)`.

The solver returns the unknown currents; validation comes next.

Reconstruct the current in the shared branch



Mesh currents from solution

$$I_1 = 2\,\text{A}, \quad I_2 = 1\,\text{A}$$

Shared branch current

$$I_{\text{shared}} = I_1 - I_2$$
$$= 2\,\text{A} - 1\,\text{A}$$
$$I_{\text{shared}} = 1\,\text{A}$$

KEY POINTS

- The shared resistor carries the mesh-current difference.
- $I_{\text{shared_A}} = I_{1_A} - I_{2_A}$.
- $I_{\text{shared_A}} = 1\,\text{A}$.
- A negative value reverses assigned direction.

Validate the original equations directly

1 Reconstruct the left and right equations

Rebuild both KVL values.

$$3(2) - 1(1) = 5\,\mathrm{V} \qquad -1(2) + 4(1) = 2\,\mathrm{V}$$

2 Check the computed currents in MATLAB

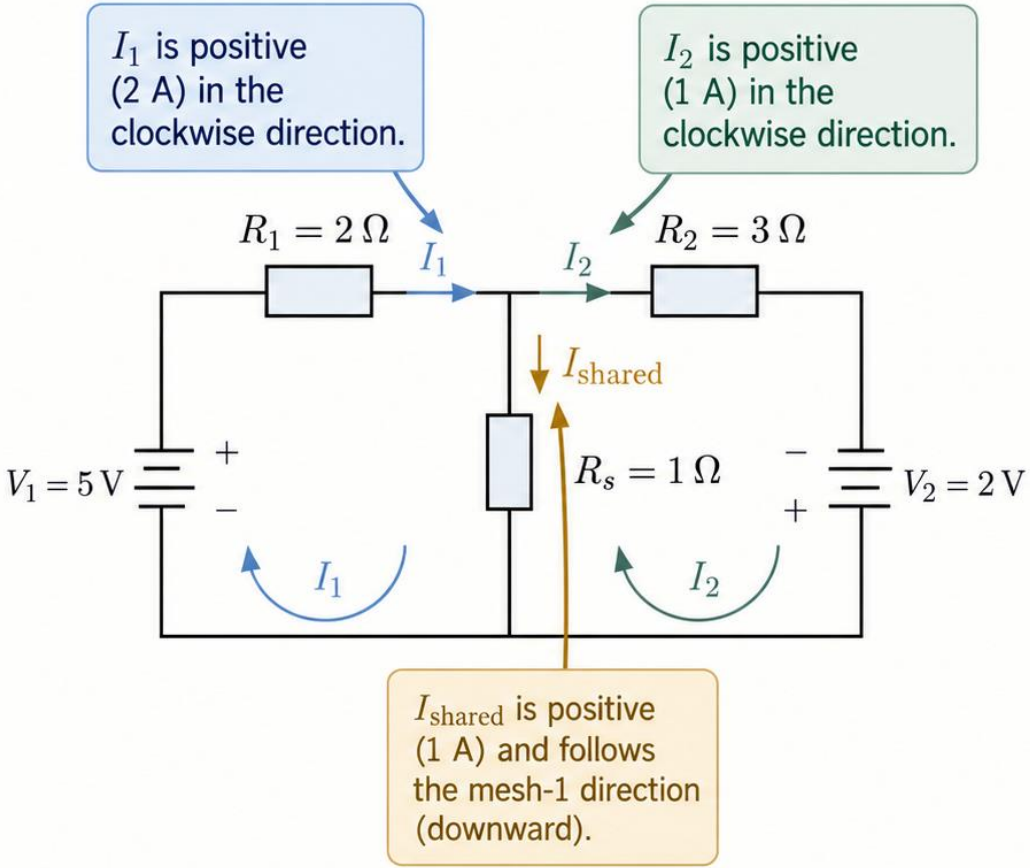
```
left_reconstructed_V = (R1_ohm + Rs_ohm)*I1_A - Rs_ohm*I2_A;
right_reconstructed_V = -Rs_ohm*I1_A + (R2_ohm + Rs_ohm)*I2_A;
validation_difference_V = [left_reconstructed_V - V1_V; ...
                           right_reconstructed_V - V2_V];
assert(max(abs(validation_difference_V)) < 1e-12, ...
    'Direct-substitution validation failed.')
```

Difference vector is zero within tolerance.

difference vector = [0; 0] V

accepted model

Read the numbers back into the circuit



Interpret the results

1. Positive I_1 and I_2 follow clockwise arrows.
2. $I_1 > I_2$ matches prediction.
3. Shared current follows mesh-1 direction.
4. Negative current means reversed physical direction.

Solved currents

$$I_1 = 2\,\text{A}, \quad I_2 = 1\,\text{A}, \quad I_{\text{shared}} = 1\,\text{A}$$

$$I_1 > 0, \quad I_2 > 0, \quad I_1 > I_2$$

Signs and magnitudes confirm the physical picture.

A plausible matrix can still encode the wrong circuit

WRONG SIGN

Matrix with a flipped off-diagonal coefficient (plausible but incorrect):

```
A_wrong_sign_ohm = [3, 1; ...
                    -1, 4];

x_wrong_sign_A = [1.3846; 0.8462];
```

FINITE OUTPUT

MATLAB runs without error and produces a finite solution:

```
>> x_wrong_sign_A
ans =
    1.3846
    0.8462
```

PHYSICS CHECK

Direct substitution into the correct equations shows a mismatch:

$$A_{\mathrm{ohm}} x_{\mathrm{wrong_sign_A}} - b_{\mathrm{V}} = \begin{bmatrix} -1.6923 \\ 0 \end{bmatrix} \mathrm{V}$$

$$\text{correct left-equation mismatch} = -1.6923\ \mathrm{V}$$

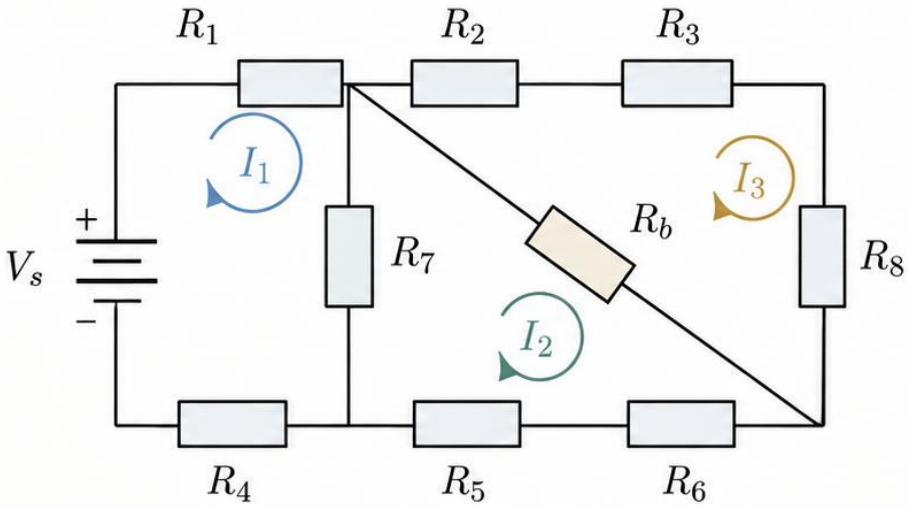
KEY POINTS

- Flipping one off-diagonal sign still gives finite output.
- Wrong matrix changes shared-resistor term.
- Correct equations expose −1.6923 V mismatch.
- Successful execution is not proof.

repair cue: revisit current direction and KVL sign

Check the shared-branch current direction and the sign of the shared-resistor term in your KVL equations.

Future exposure — the same pattern scales to a bridge circuit



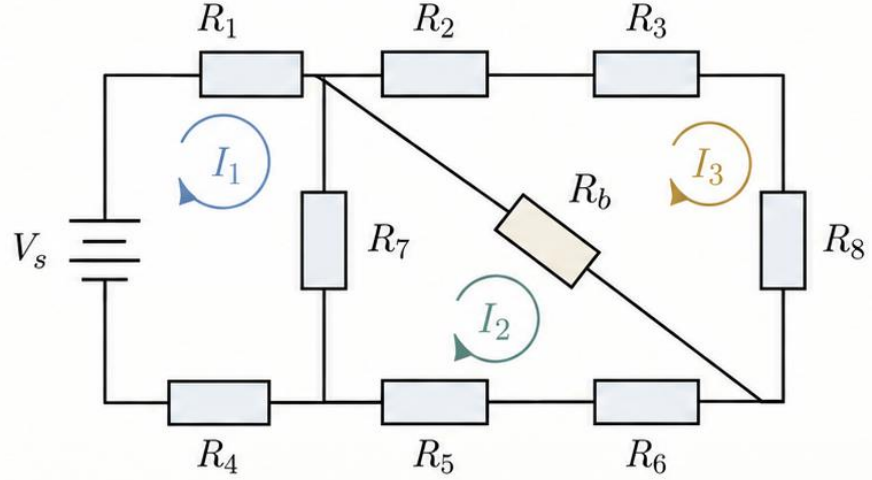
Optional — future exposure

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}}_{\text{coefficient matrix}} \underbrace{\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}}_{\text{unknown mesh currents}} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}}_{\text{source terms}}$$

KEY POINTS

- Three clockwise mesh currents I_1, I_2, and I_3 lead to three Kirchhoff equations.
- R_7 couples I_1-I_2; diagonal R_b couples I_2-I_3; I_1 and I_3 share no branch.
- Rows are equations; columns are mesh-current unknowns.
- MATLAB can solve a 3×3 $A*x = b$ after assembly, but no solution is shown here.

Self-activity — sketch the bridge matrix before MATLAB



$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}}_{\text{coefficient matrix}} \underbrace{\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}}_{\substack{\text{unknown} \\ \text{mesh currents}}} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}}_{\text{source terms}}$$

1 Mark I_1, I_2, I_3 and list shared branches R_7 and R_b.

2 Predict matrix dimension and row/column meaning.

3 Separate self-resistance from shared-branch entries.

4 Sketch one validation check.

5 Stop at scaffold; no coefficients required today.

Optional future exposure | **Topology:** R_7: I_1–I_2 shared | R_b: I_2–I_3 shared | I_1 and I_3: no shared branch

Working exposure — separate representation changes from physical input changes

Representation change

Swap A and b together → same solution

$$A\mathbf{x} = \mathbf{b}$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

Reorder A and b together

Same physical solution

$$PAP^T\mathbf{x}' = P\mathbf{b}$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} I_2 \\ I_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

```
maximum_reordering_difference_A = 0
```

Physical input change

Change V_2 → both coupled currents change

V_2: 2.0 V ⟶ 2.1 V



$$\mathbf{x}_{\text{perturbed}} = \begin{bmatrix} 2.0091 \\ 1.0273 \end{bmatrix} \text{A}$$

$$\Delta\mathbf{x} = \begin{bmatrix} 0.0091 \\ 0.0273 \end{bmatrix} \text{A}$$

Ask: did the physics change, or only the representation?

Stretch checks — optional

Residual • rank • conditioning • power balance

Exit ticket — carry the model-to-check chain forward

PREDICT

state unknowns, directions, units.

MAP

trace one coefficient to its KVL term.

CHECK

write one direct-substitution validation.

INTERPRET

explain a negative current and one limitation.

Transfer cue: model → equations → solve → validate → interpret